Sampling and Data

Questions 1–114

1.47 — Question 1. A by-election is due to take place in Wellingborough. Amelia is a politics student who is interested in the election and decides to survey local residents. She stands in the main high street and surveys the first 500 people she meets. Identify the name of this type of sampling.

simple random opportunity
cluster quota

1.47 — Question 2. In a small village of 156 households, Jason wants to investigate how many use the local post office. He decides to number the households from 001 to 156, and then use a random number generator to obtain 30 different three-digit numbers between 001 and 156. He will then select the corresponding households and include them in the sample. Identify the name of this type of sampling.

convenience self-selection
simple random systematic

1.47 — Question 3. The table below shows the share price of HSBC on the 1st January, 1st April, 1st July and 1st October across a sample of five years.

	01/01	01/04	01/07	01/10
1993	156.57	173.13	189.10	221.34
1998	464.77	548.13	442.40	406.67
2003	549.88	597.37	671.01	771.22
2008	654.88	766.43	731.13	664.03
2013	716.70	703.50	747.80	682.10

Identify the sampling procedure that most likely chose the years selected in the sample.

quota systematic
simple random stratified

1.47 — Question 4. In a secondary school, there are 12 classes in year 9 and the school wants to collect some feedback on a recent futures fair that visited the school. Sandra, the head of year 9, decides to select three classes at random and then get feedback from every student in those classes. Identify the name of this type of sampling.

stratified quota
opportunity cluster

1.47 — Question 5. Ibrahim works for a local charity and is asked to interview members of the public in the high street on a Saturday afternoon. He is asked to interview 30 people under the age of 40, and 30 people who are aged 40 or over. Identify the name of this type of sampling.

simple random stratified
systematic quota

1.47 — Question 6. A large retailer of flat-pack furniture employs 7200 members of staff across 24 stores in the UK. The retailer wants to investigate their staff's job satisfaction through a survey. The HR team suggest four possible sampling methods:

Method A Randomly select 120 people from the whole list.

Method B Randomly select five stores and then randomly select 24 people from each of these stores.

Method C Randomly select 120 people from the London store.

a) Identify the population in the retailer's investigation.

b) Give one advantage of using Method B over Method C.

c) Give one advantage of using Method B over Method A.

1.47 — Question 7. The Office for National Statistics conducts a census for England and Wales every 10 years. Describe one disadvantage with a census.

1.47 — Question 8. In 2019, there were estimated to be approximately 20,850 breeding pairs of African penguins. The health of the population is to be investigated by sampling 100 breeding pairs. Explain why a simple random sample cannot be used.

1.47 — Question 9. A national union of care workers is holding its annual conference. Chloe wants to sample the views of care workers on conditions in their place of work. She decides to set up a stall near the entrance to the conference and ask attendees who visit her stall to fill in a questionnaire. State two disadvantages with this sampling method.

1.47 — Question 10. Clive runs a takeaway business. He wants to revise the menu down to reduce costs. He has recorded the choices made by each customer since the business opened and now has 680 customers and their choices in a spreadsheet. Describe how Clive could obtain a simple random sample of size 40 from his list of customers.

1.47 — Question 11. Lundi supermarkets split their workforce into those who work on the shop floor and those who work in the warehouse. Across all of their stores, 3265 work in the shops and 2435 work in the warehouses. The company wants to conduct a survey of 150 members of staff, and they decide to use a stratified sample. Calculate how many from the shop floor and how many from the warehouses should be included in the sample.

1.47 — Question 12. Harry works in a large office block. Near his office, there is a kitchen where employees store their mug in a cupboard. The mugs are all different, but his is often used by somebody else accidentally. Harry wants to label all the mugs with their owner's name. Suggest why an opportunity sample conducted near the kitchen would be more appropriate to gather opinions than a simple random sample to the whole workforce.

1.47 — Question 13. Susan is the Subject Leader of Maths at a large sixth form college. She wants to investigate the number of hours it takes for the students taking A-Level Maths to complete their homework each week. There are 1200 students in 50 classes. Describe how Susan could obtain a cluster sample of three classes of students for her survey.

1.47 — Question 14. A company wishes to survey its employees. It decides to send out an online survey to all of their employees' work emails and then analyse the results they get within a 24-hour window. Suggest two disadvantages with using this method.

1.47 — Question 15. A work-place choir is made up of four companies: Aurora, BrightByte, Cosmic and Dreamscape. The number of people from each company who take part in the choir are 25, 30, 50 and 45 respectively. A stratified sample is taken of size k to gain feedback on the choir, and the samples taken from Aurora, BrightByte and Cosmic are 8, 10 and 17 respectively. In the calculations to determine how many people must be sampled from each company, the number was always rounded to the nearest person.

a) Determine the value of k

b) Determine how many people who work at Dreamscape should be sampled.

1.47 — Question 16. A sixth form college employs 120 members of staff. A survey is to be sent out to 20 members of staff using the following method:

- List all of the members of staff in alphabetical order by surname and assign each of them a number from 1 to 120.
- Choose a random number from 1 to 6 and choose this person.
- Select every 6th person on the list from there until 20 people are selected.

a) Identify this sampling method.

b) Give a reason why this method does not provide a random sample.

1.48 — Question 1. For the two data sets below, find the mean, median, mode, range and interquartile range, and hence compare the two sets of data.

Set A	3	3	6	7	8	12	15	16	20	25	28
Set B	3	3	6	9	10	12	13	16	20	23	28

1.48 — Question 2. Give one reason why:

a) the mean would be preferable to use as an average instead of the median,

b) the median would be preferable to use as an average instead of the mean.

1.48 — Question 3. The following data was recorded:

1	4	8	8	9	10	11	14	15	15	25

a) Find the mean and standard deviation.
b) Values that are less than two standard deviations below or above the mean are considered outliers. Determine whether there are any outliers in the data.
c) Find the median and interquartile range.
d) Values that are less than LQ $-1.5\times$ IQR or more than UQ $+1.5\times$ IQR are considered outliers. Determine whether there are any outliers in the data.

1.48 — Question 4. The following data was recorded:

3	8	9	10	15	17	21	25	26	28	29	55

a) Find the mean and standard deviation.
b) Values that are less than two standard deviations below or above the mean are considered outliers. Determine whether there are any outliers in the data.
c) Find the median and interquartile range.
d) Values that are less than LQ $-1.5\times$ IQR or more than UQ $+1.5\times$ IQR are considered outliers. Determine whether there are any outliers in the data.

1.48 — Question 5. The following data was recorded:

35	36	45	48	52	59

a) Find the mean.
b) Without performing any calculations, explain what would happen to the mean if 59 was removed.

1.48 — Question 6. The following data was recorded:

21	25	40	61	90	102

a) Find the mean.
b) Without performing any calculations, explain what would happen to the mean if 40 was changed to 56.5

1.48 — Question 7. Find the mean, $\bar{x}$,

a) given that $\sum x = 860,\ n = 5$

b) given that $\sum x = 86753,\ n = 35$

1.48 — Question 8. A data set, x, contains n values. Given that $\sum x = 51324$ and $\bar{x} = 52$, determine the value of n

1.48 — Question 9. Given that $\sum x = 613, \sum x^2 = 56905, n = 10$, find the standard deviation, σ, to 3 significant figures.

1.48 — Question 10. Given that $\sum x = 412, \sum x^2 = 7322, n = 25$, find the standard deviation, σ, to 3 significant figures.

1.48 — Question 11. Given that $\sum x = 3305, \sum x^2 = 183075, n = 60$, find the variance, σ^2, to 3 significant figures.

1.48 — Question 12. Given that $\sum x = 1354, \sum x^2 = 212182, n = 9$, find the variance, σ^2, to 3sf.

1.48 — Question 13. The following data was recorded:

109	112	113	117	118	125

a) Find the standard deviation.
b) Without performing any calculations, explain what would happen to the standard deviation if 125 was removed.

1.48 — Question 14. The following data was recorded:

37.4	39.2	40.1	40.3	42.1	43.6

a) Find the standard deviation.
b) Without performing any calculations, explain what would happen to the standard deviation if 43.6 was changed to 48.6

1.48 — Question 15. The following data was recorded:

8	11	13	16	18	21

a) Find the mean.
b) Find the standard deviation.
c) Two more data points are added, x and y, such that $x < y$. It is given that the mean remains the same but the standard deviation is reduced. Determine possible values for x and y

1.48 — Question 16. The following data was recorded:

21	25	32	34	38	42

a) Find the mean.
b) Find the median.
c) Two more data points are added, x and y, such that $x < y$. It is given that the mean remains the same but the median is reduced. Determine possible values for x and y

1.48 — Question 17. The following data was recorded:

8	12	15	x	22	28

a) Given that the mode is 22, find the value of x
b) Given instead that the mean is 17.05, find the value of x
c) Given instead that the median is 16.3 and $15 < x < 22$, find the value of x

1.48 — Question 18. Chrissy grows a certain variety of sunflowers. One day, she measures the heights, in cm, of a random sample of 12 of the sunflowers in her garden and her collected data is shown below.

85	102	121	127	132	133

140	145	149	160	162	198

a) Find the mean height of the sunflowers to 3 significant figures.
b) Find the sample standard deviation of the heights of the sunflowers to 3 significant figures.
c) Any sunflower with a height of less or more than 2 standard deviations from the mean may be regarded as an outlier. Using your answers to a) and b), determine if there are any outliers.

1.48 — Question 19. Collin sampled 80 students at his school about the number of pets their household had. This data is shown in the following table:

Number of pets	Frequency
0	7
1	31
2	21
3	8
4	6
5	5
6	2

Calculate the mean number of pets per household, and the standard deviation.

1.48 — Question 20. Alexis surveyed 95 colleagues at her company about the number of projects they were working on. This data is shown in the following table:

Number of projects	Frequency
0	3
1	34
2	48
3	10

Calculate the mean number of projects per employee, and the standard deviation.

1.48 — Question 21. Sofia sampled 60 properties listed on an estate agent's website and recorded the number of bedrooms the properties had. This data is shown in the following table:

Number of bedrooms	Frequency
1	12
2	18
3	21
4	6
5 to 7	2
8 to 10	1

a) Calculate an estimate for the mean number of bedrooms.
b) Find the median number of bedrooms.

1.48 — Question 22. The following data was recorded:

x	Frequency
$14 \leq x < 18$	100
$18 \leq x < 22$	200
$22 \leq x < 30$	250
$30 \leq x \leq 32$	50

a) Calculate an estimate for the mean of x
b) Calculate an estimate for the standard deviation of x
c) State the assumption that explains why the midpoints of the class intervals are used to estimate the mean and standard deviation.

1.48 — Question 23. The weights, in kg, of 30 swans kept on a nature reserve was recorded:

Weight	Frequency
5 to 8	5
9 to 11	15
12 to 13	8
14 to 15	2

a) State the upper class boundary of the “9 to 11" class.
b) Calculate an estimate for the mean weight of the swans.
c) Calculate an estimate for the standard deviation of the weights of the swans.

1.48 — Question 24. The ages, in years, of 90 teachers working at a sixth form college was recorded:

Age	Frequency
22 to 26	12
27 to 34	30
35 to 45	22
46 to 54	19
55 and over	7

a) State the upper class boundary of the “22 to 26" class.
b) Explain why changing the “55 and over" class to “55 to 66" would be reasonable.
c) Using the class interval suggested in b), calculate estimates for the mean and standard deviation of the ages of the teachers.

1.48 — Question 25. A publishing company based in London employs 17 members of staff. Their salaries, in thousands of pounds, are recorded in the table below.

Salary	Frequency
35 –	3
40 –	7
45 –	4
50 –	1
55 – 75	2

a) Estimate the mean of the company's salaries.
b) Estimate the standard deviation of the company's salaries.
c) Explain why your answers to a) and b) are only estimates.
d) Explain why your answer to a) should not be publicised as the average wage of people who work in publishing across the UK.

1.48 — Question 26. Given that $\bar{x} = 9, \sigma^2 = 2, n = 33$, find $\sum x$ and $\sum x^2$

1.48 — Question 27. Given that $\bar{x} = 90.2, \sigma^2 = 10.96, n = 100$, find $\sum x$ and $\sum x^2$

1.48 — Question 28. Ebba takes a contract with a company that makes A-Level Maths resources. Her contract says that the average number of hours working per week should be approximately 4, although they will pay her for each hour worked up to a maximum of 8 hours a week. Some projects take Ebba longer than this. Ebba records the number of hours, t, she works each week in the table below.

Hours Worked (t)	Frequency
$0 \leq t < 2$	12
$2 \leq t < 4$	17
$4 \leq t < 6$	21
$6 \leq t < 8$	7
$8 \leq t < 12$	2

Ebba is paid £15.60 an hour for her work. Estimate the mean gross pay she earns from the company each week.

1.48 — Question 29. Given that $\sum x = 40000, \sum x^2 = 80000, \sigma = 1$, find n

1.48 — Question 30. The following data has a mean of 30 and a sample standard deviation of 8.70 to 3 significant figures.

19	21	26	27	29	35	39	44

One piece of data is removed and replaced with a new value. The mean increases to 32 but the sample standard deviation decreases. Identify which of the pieces of data could have been removed and find the values they would need to be replaced with.

1.48 — Question 31. It is given that $\bar{x} = k, \sigma = 5$ and $n = 10$. Two more pieces of data are added, $x = k - 1$ and $k + 1$. Determine the standard deviation of the 12 pieces of data.

1.48 — Question 32. It is given that $\bar{x} = 200, \sigma = k$ and $n = 10$. Two more pieces of data are added, $x = 2 + k$ and $2 - k$ so that σ is now 99. Determine the value of k

1.48 — Question 33. On a rainy day, 23 students sit a test. Their teacher marks their papers and finds there to be a mean score of 30 and a standard deviation of 5. Due to the rain, Tom was late to school and missed the test, and so had to complete it later. Tom scored 35. Determine the mean and standard deviation of the scores of all 24 students.

1.48 — Question 34. On a sunny day, 25 students sit a test. Their teacher marks their papers and finds there to be a mean score of 40 and a standard deviation of 8. The teacher finds that one of his students has cheated and copied another's work. The student's score was 52 and the teacher wants to remove this score from the analysis. Determine the mean and standard deviation of the scores of these remaining 24 students.

1.48 — Question 35. Linear Coding The data set, x, shown below is coded using $y = \frac{x+5}{4}$

x	15	85	155	295	455

Write down the data set y

1.48 — Question 36. The values x_i have a mean of $\bar{x} = 85$ and standard deviation $\sigma_x = 12$. The data is coded to the values y_i such that $y = 10x$
Find $\bar{y}$ and σ_y, the mean and standard deviation respectively of the values y_i

1.48 — Question 37. The values x_i have a mean of $\bar{x} = 120$ and standard deviation $\sigma_x = 5$. The data is coded to the values y_i such that $y = 3x + 50$
Find $\bar{y}$ and σ_y, the mean and standard deviation respectively of the values y_i

1.48 — Question 38. The values x_i have a mean of $\bar{x}$ and standard deviation σ_x. The data is coded to the values y_i such that $y = 8x - 10$. It is given that $\bar{y} = 500$ and $\sigma_y = 120$
Find $\bar{x}$ and standard deviation σ_x

1.48 — Question 39. The values x_i have a mean of $\bar{x}$ and standard deviation σ_x. The data is coded to the values y_i such that $y = \frac{20-3x}{9}$. It is given that $\bar{y} = 250$ and $\sigma_y = 45$. Find $\bar{x}$ and standard deviation σ_x

1.48 — Question 40. The temperatures (°C) at midday for a week in May in Valletta, Malta are recorded in the table below.

24	23	25	27	29	26	24

a) Find the mean and standard deviation of the temperatures in °C
b) The formula to convert from °C to °F is given by $F = \frac{9C}{5} + 32$
Use this formula to determine the mean and standard deviation of the temperatures in °F

1.48 — Question 41. The temperatures (°F) at midday for a week in May in Southampton, UK are recorded in the table below.

61	63	59	59	64	63	61

a) Find the mean and standard deviation of the temperatures in °F

b) The formula to convert from °F to °C is given by $C = \frac{5(F-32)}{9}$
Use this formula to determine the mean and standard deviation of the temperatures in °C

1.48 — Question 42. The Mid-Range
For the following data, find the mid-range.

28	35	42	59	60	75

1.48 — Question 43. Colm teaches Biology to four classes. He set some homework for his students on an online platform and his dashboard recorded how often his students logged in during the week to work on it. This data is shown in the following table:

Log-ins	Frequency
0	12
1	42
2	27
3	10
4	8
5	3
6	1

a) State the mode of the data.
b) State the mid-range of the data.
c) State the median of the data.

1.48 — Question 44. Linear Interpolation
The following data was recorded:

x	Frequency
$0 \leq x < 5$	35
$5 \leq x < 10$	60
$10 \leq x < 15$	15
$15 \leq x \leq 20$	10

Use linear interpolation to calculate an estimate for the median of x

1.48 — Question 45. The following data was recorded:

x	Frequency
$0 \leq x < 8$	82
$8 \leq x < 12$	74
$12 \leq x < 20$	62
$20 \leq x \leq 24$	44

Use linear interpolation to calculate an estimate for the median of x

1.48 — Question 46. The following data was recorded:

x	Frequency
$10 \leq x < 30$	55
$30 \leq x < 70$	95
$70 \leq x < 80$	150
$80 \leq x < 110$	75
$110 \leq x \leq 140$	25

Use linear interpolation to calculate an estimate for the median of x

1.48 — Question 47. The following data was recorded:

x	Frequency
$50 \leq x < 60$	12
$60 \leq x < 70$	16
$70 \leq x < 80$	28
$80 \leq x < 100$	32
$100 \leq x \leq 120$	14

Use linear interpolation to calculate an estimate for the interquartile range of x

1.48 — Question 48. The following data was recorded:

x	Frequency
$0 \leq x < 15$	48
$15 \leq x < 40$	93
$40 \leq x < 55$	117
$55 \leq x < 70$	152
$70 \leq x \leq 95$	146
$95 \leq x < 110$	121
$110 \leq x < 135$	94
$135 \leq x < 150$	29

a) Use linear interpolation to calculate an estimate for 20th to the 80th interpercentile range of x
b) Estimate the number of data values that fall within the 20th to the 80th interpercentile range.

1.48 — Question 49. Andrew measures the acidity of the soil in several randomly selected locations at a nursery. The following data was recorded:

x (pH)	Frequency
$6 \leq x < 6.2$	12
$6.2 \leq x < 6.4$	27
$6.4 \leq x < 6.6$	38
$6.6 \leq x < 6.8$	22
$6.8 \leq x < 7.0$	9

Use linear interpolation to calculate an estimate for the upper quartile of the pH levels recorded.

1.48 — Question 50. Freya measures the heights, in metres, of a random sample of oak trees in a larger forest. The following data was recorded:

Height (m)	Frequency
13 to 15	7
16 to 18	15
19 to 21	23
22 to 24	9
25 to 26	2

Use linear interpolation to calculate an estimate for the lower quartile of the heights of the oak trees recorded.

1.48 — Question 51. Lisa runs a carp farm that has over 50 outdoor ponds dedicated to stocking fisheries in the UK. In one pond, there are a large number of young carp. Lisa collects a sample and measure their lengths in centimetres. The following data was recorded:

Length (cm)	Frequency
6 to 7	56
8 to 9	84
10 to 11	129
12 to 13	76
14 to 15	43

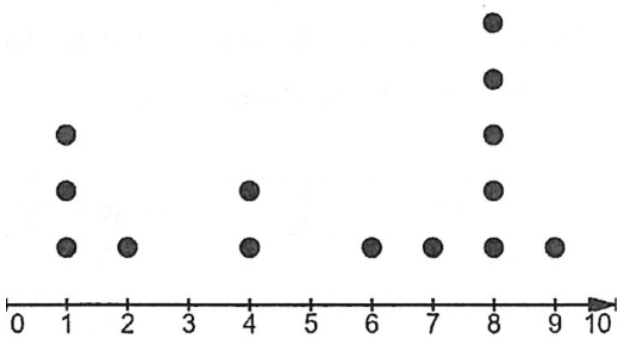
Use linear interpolation to calculate an estimate for the length 70% of the fish are longer than.

1.48 — Question 52. Gerald records the ages, in years, of the members of a local choir. The following data was recorded:

Age (years)	Frequency
25 to 30	9
31 to 40	32
41 to 50	19
51 to 65	26
66 to 80	12

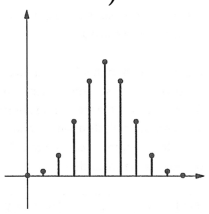
Use linear interpolation to calculate an estimate for the median age of the choir.

1.49 — Question 1. Find the mean from the dotplot:

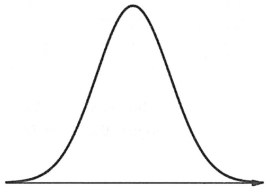


1.49 — Question 2. Use the terms: uniform, unimodal, bimodal, continuous, discrete, symmetric, positively skewed, and/or negatively skewed to describe:

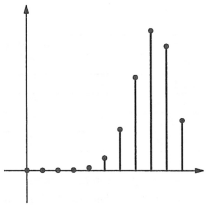
a)



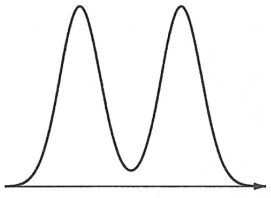
b)



c)

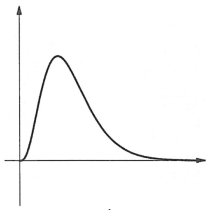


d)

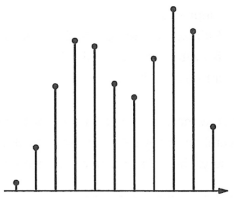


1.49 — Question 2 (continued). Use the terms: uniform, unimodal, bimodal, continuous, discrete, symmetric, positively skewed, and/or negatively skewed to describe:

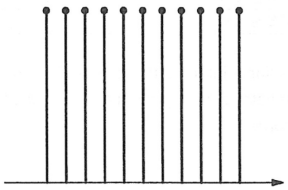
e)



f)



g)



1.49 — Question 3. The stem-and-leaf diagram shows the weights of a sample of 29 male moose measured to the nearest kg.

38	1	2	5	6	6			
39	3	3	4	7	8	9	9	9
40	0	0	2	2	3	8		
41	0	1	2	3	9			
42	2	2	2					
43	0	1						

Key: 38|1 represents 381 kg

a) Describe the shape of the distribution.
b) Find the median.
c) Find the interquartile range.

1.49 — Question 4. The stem-and-leaf diagram shows the heights of a sample of 13 red kangaroos measured to the nearest cm.

7	2 3 3
8	1 2 3 6 7 8
9	5 5
10	0
11	
12	9

Key: 7|4 represents 74 cm

a) Find the median and interquartile range of the red kangaroo heights.
b) Calculate the mean and the standard deviation of the red kangaroo heights.
c) Explain why the median would be a better measure of central tendency than the mean in this case.

1.49 — Question 5. The stem-and-leaf diagram shows the wingspans of a sample of 23 Arctic terns measured to the nearest cm.

31	0
32	8 9
33	0 1 2 4
34	0 0 4 4 8
35	3 3 7 8 8 9
36	0 0 0 1 1

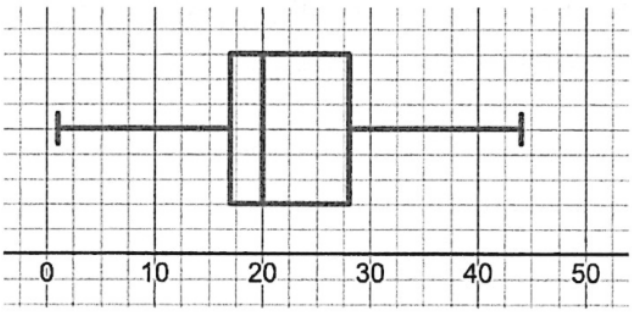
Key: 31|0 represents 31.0 cm

a) Describe the shape of the distribution.
b) Alan looks at the data in the stem-and-leaf diagram and determines that 36.0 cm is the average wingspan of an Arctic tern. Explain how Alan can justify this result.
c) Brian looks at the data in the stem-and-leaf diagram and determines that 34.8 cm is the average wingspan of an Arctic tern. Explain how Brian can justify this result.
d) Draw a boxplot to represent the data.

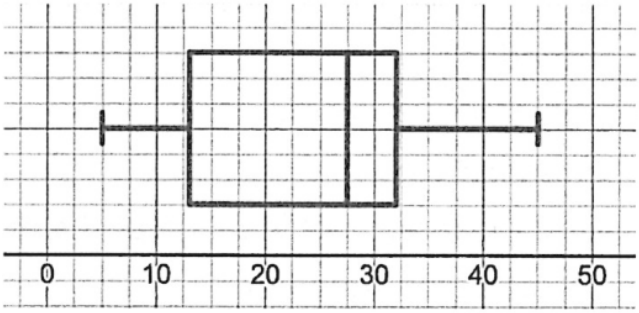
1.49 — Question 6. Draw a boxplot with negative skew.

1.49 — Question 7. For the following boxplots, identify whether they are roughly symmetric, positively skewed or negatively skewed:

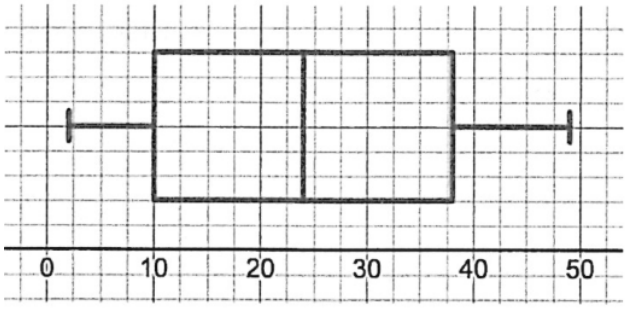
a)



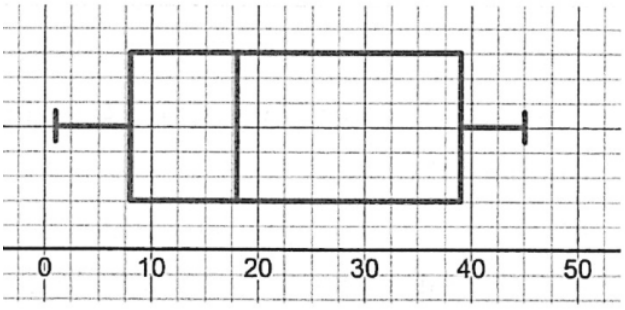
b)



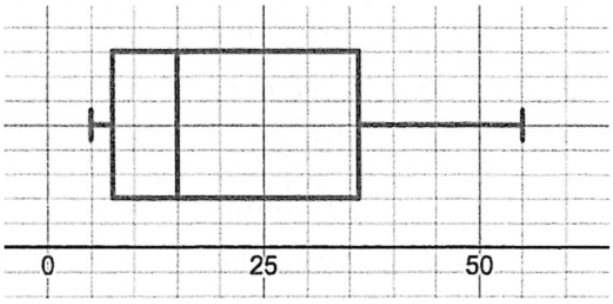
c)



d)

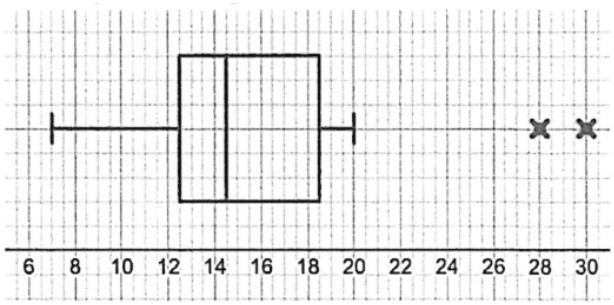


1.49 — Question 8. The boxplot below represents the commute times, in minutes, of employees at a large company.



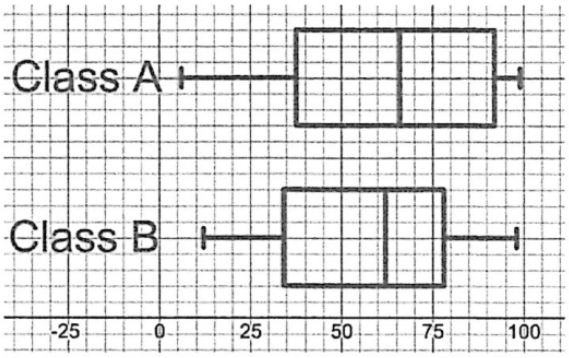
a) State the name of the measure of central tendency identified by the box plot, and its value.
b) Determine whether the box plot is symmetric, negatively skewed or positively skewed.

1.49 — Question 9. The boxplot below represents the number of dogs that attended puppy training each Thursday evening for 20 weeks.



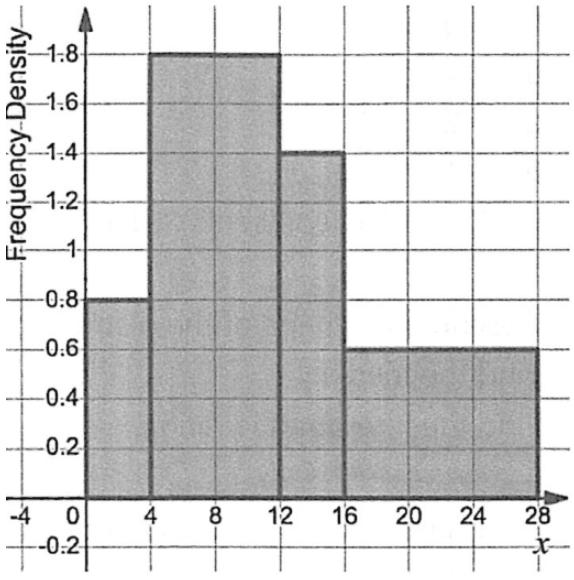
a) Find the median number of dogs that attended.
b) Find the range of the number of dogs that attended.
c) Find the interquartile range of the number of dogs that attended.

1.49 — Question 10. Two classes sat a test where the maximum score was 100. The results are displayed in the two boxplots below.



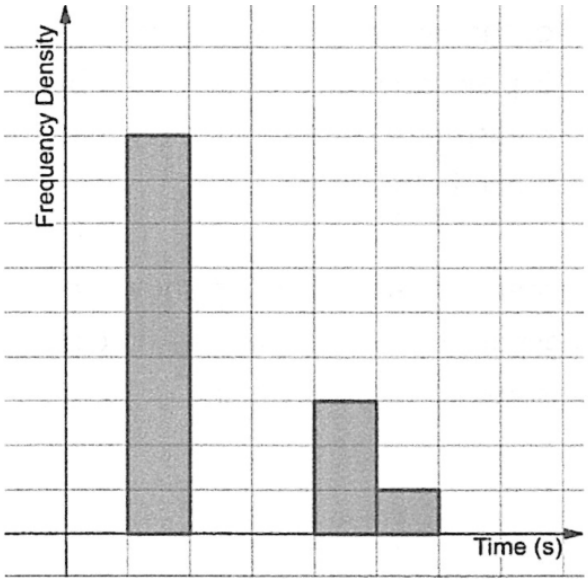
a) Estimate the median of Class A and the median of Class B.
b) Estimate the interquartile range of Class A and the interquartile range of Class B.
c) Use your answers to compare the results of the two classes.

1.49 — Question 11. The histogram below shows the times, x minutes, that employees at a company take to walk to work. Estimate the proportion of employees that take between 9 minutes and 25 minutes to walk to work.



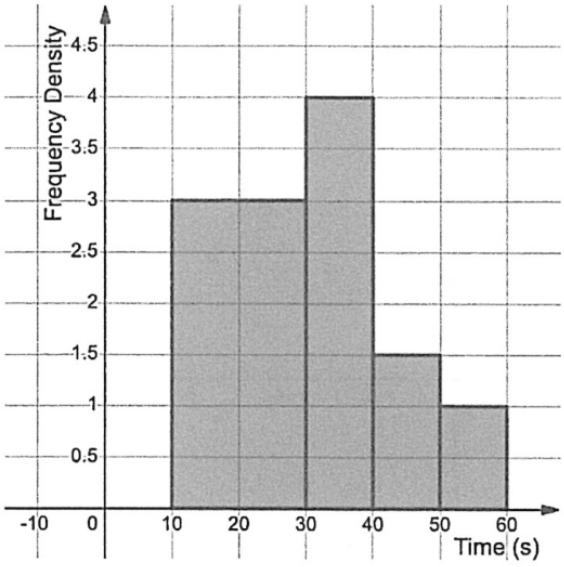
1.49 — Question 12. Aminah is investigating the length of time, t minutes, meetings take at her place of work. She is partway through drawing a histogram from the following table of data:

Meeting Length (t)	Frequency
$0 \leq t < 30$	30
$30 \leq t < 60$	45
$60 \leq t < 120$	20
$120 \leq t < 150$	15
$150 \leq t < 180$	



a) Copy and complete the table and histogram.
b) Estimate the number of meetings that last between 40 and 99 minutes.

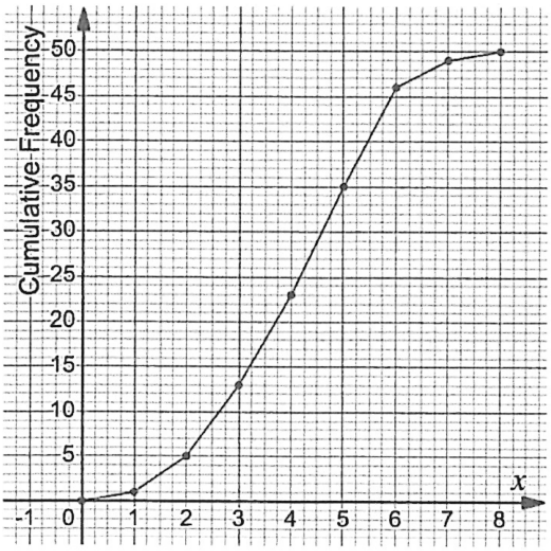
1.49 — Question 13. The histogram below shows the times, in seconds, for a sample of adults to complete a puzzle.



a) Which class interval contains the median?
b) Estimate the mean.
c) Estimate the standard deviation to 3sf.
d) An adult is selected at random. Find the probability that they took between 28 and 32 seconds to complete the puzzle.

1.49 — Question 14. Umar took a random sample of 24 apples and they weighed between 160 and 180 grams. He drew a histogram to represent the data and the bar representing 160 to 180 grams has a width of 2 cm and a height of 6 cm. There are 16 apples that weigh between 180 and 210 grams. Determine the width and height of the bar on the histogram representing 180 to 210 grams.

1.49 — Question 15. Use the following cumulative frequency graph to estimate the median and interquartile range.

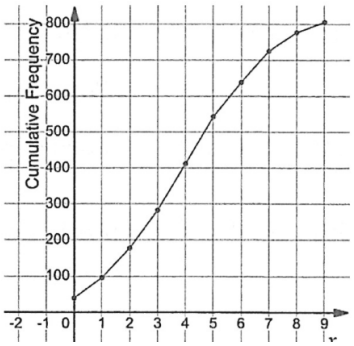


1.49 — Question 16. Amber sells ice-creams. One year, she counts the number of customers she sells ice creams to over ten weeks of Sundays in the summer. Her sales are shown in the table below:

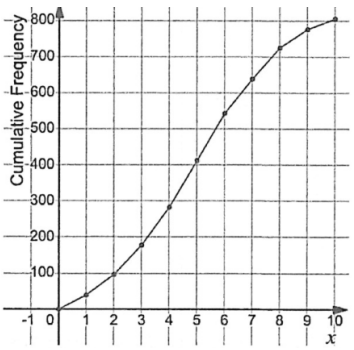
Week	0 –	1 –	2 –	3 –	4 –
Sales	40	56	82	105	129
Week	5 –	6 –	7 –	8 –	9 –
Sales	131	95	87	51	29

a) Amber draws two different cumulative frequency graphs. These are shown below. Which of the two should she use to represent her data? Explain your reasoning.

Graph A:



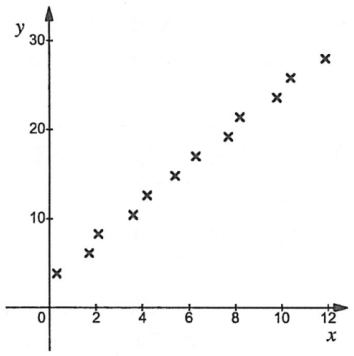
1.49 — Question 16 (continued). Graph B:



b) Amber shows the correct graph to Terry who says that it shows that week on week her sales are increasing. With reference to the graph, explain whether Terry is correct.

c) Terry goes on to say that Amber can use the correct graph to estimate the exact time during a week that Amber had sold half of the ice creams. Explain why this would be inappropriate.

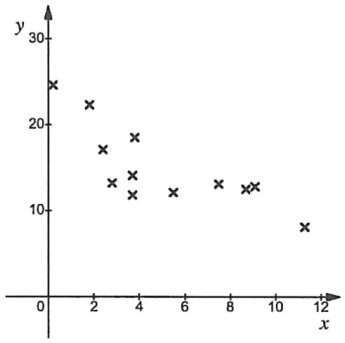
1.50 — Question 1. The diagram below shows bivariate data.



Which of the following best describes the correlation shown in the diagram?

Strong Positive	Moderate Positive
Weak Positive	Weak Negative
Moderate Negative	Strong Negative

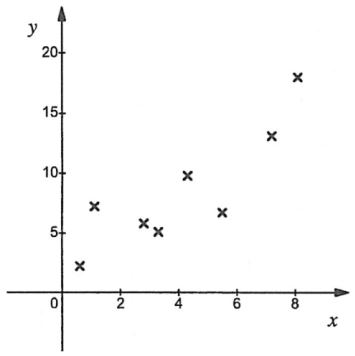
1.50 — Question 2. The diagram below shows bivariate data.



Which of the following best describes the correlation shown in the diagram?

Strong Positive	Moderate Positive
Weak Positive	Weak Negative
Moderate Negative	Strong Negative

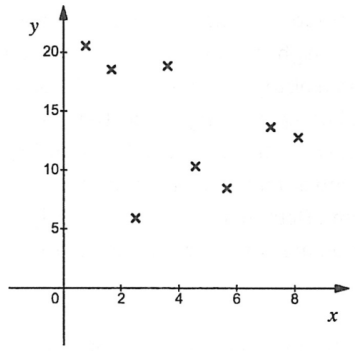
1.50 — Question 3. The diagram below shows bivariate data.



One of the following product moment correlation coefficient is correct for the above data. Which is it?

$$\begin{array}{cc} 0.864 & 0.164 \\ -0.164 & -0.864 \end{array}$$

1.50 — Question 4. The diagram below shows bivariate data.



One of the following product moment correlation coefficient is correct for the above data. Which is it?

$$\begin{array}{cc} 0.891 & 0.391 \\ -0.391 & -0.891 \end{array}$$

1.50 — Question 5. Which of the following cannot be a value for the product moment correlation coefficient?

$$0 \qquad \frac{3}{4} \qquad -\frac{7}{3} \qquad -1 \qquad \frac{8}{7}$$

1.50 — Question 6. A researcher conducts a study to understand the impact of physical exercise on mental health among college students. The researcher collects data on the number of hours students exercise per week and their levels of stress, measured on a standardised scale. Identify which of these is the explanatory variable and which is the response variable:

Age of students	Hours of exercise per week
Diet quality	Levels of stress

1.50 — Question 7. Asher collects data on a person's age and their reaction time as measured in a standardised test. He calculates the product moment correlation coefficient to be 0.79. Which of the following best describes the pmcc found?

Definitely Correct Probably Correct
Probably Incorrect Definitely Incorrect

1.50 — Question 8. Jane wants to determine if there is any correlation between a person's salary and how much they spend at a large supermarket. Jane collects a random sample of adults and calculates the product moment correlation coefficient to be 1.13. Which of the following best describes the pmcc found?

Definitely Correct Probably Correct
Probably Incorrect Definitely Incorrect

1.50 — Question 9. Franklin has two data points: $(13, 7)$ and $(20, 11)$. He calculates the product moment correlation coefficient between x and y to be 1. Which of the following best describes the pmcc found?

Definitely Correct Probably Correct
Probably Incorrect Definitely Incorrect

1.50 — Question 10. Ayaan works in an office in a busy city centre. He collects a random sample of his colleagues, and his results are shown in the table below.

Distance from workplace (miles)	Commute Time (minutes)
1.2	14
2.5	21
4.1	24
5.8	31
7.3	36
9.0	39
10.7	44
12.4	51

Use your calculator to calculate the product moment correlation coefficient between the distances from the workplace and the commute times.

1.50 — Question 11. Grace asks a random sample of her the students in her year group how many hours they studied in a week (to the nearest hour) in preparation for a test, and their test score out of 100. The results are shown in the table below.

Hours studied in a week	Test Score (/100)
3	65
6	75
9	77
4	62
8	80
10	85
3	57
7	78

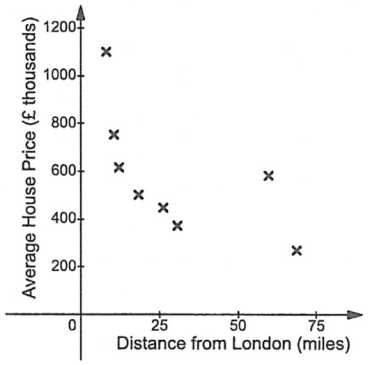
a) Use your calculator to calculate the product moment correlation coefficient between the hours studied in a week by the students and their test score.
b) Describe what the PMCC shows in this case, in context.

1.50 — Question 12. A scientist is investigating the effects of smoking on the respiratory system in youths. A random sample of 15-18 year olds found a PMCC of -0.76 between the frequency of smoking and their respiratory function. Describe what this shows in context.

1.50 — Question 13. Angela is investigating the average price of a house compared with the distance the house is away from London. She decides to consider areas in a South-Westerly direction from the centre of London. Her results are shown in the table below.

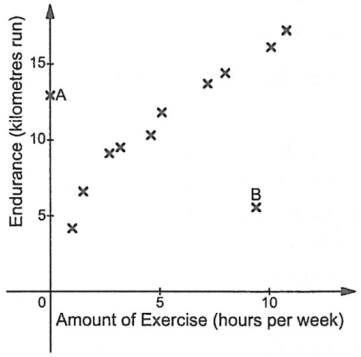
Location	Dist. from London (miles)	Av. House Price (£thousands)
Richmond	8.1	1100
Twickenham	10.5	753
Hampton	12.1	616
Chertsey	18.3	502
Bagshot	26.2	448
Farnborough	30.7	372
Winchester	59.8	582
Tidworth	68.7	269

1.50 — Question 13 (continued). Angela plotted the data in a scatter graph, as shown below.



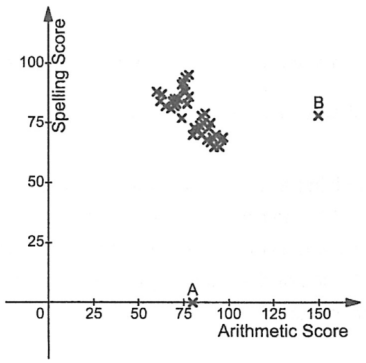
a) Identify a possible outlier and give a possible reason for it being an outlier.
b) If the outlier is discarded, determine if the scatter diagram shows any correlation for the remaining points.
c) Interpret the scatter diagram in context.

1.50 — Question 14. Charlene organises a running club. In week 10, she plots the data of her running club members with the amount of exercise they do the previous week in hours along the horizontal axis, and their endurance, measured by how far they can run on the day they meet in kilometres along the vertical axis, as shown in the graph below.



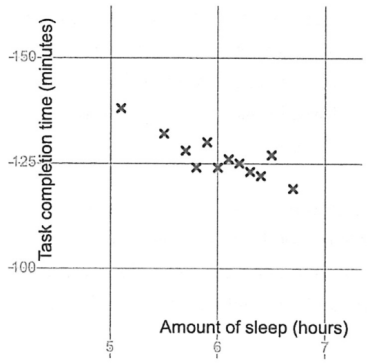
a) Explain why data point A may not have been inputted correctly.
b) Explain why data point B may have been inputted correctly.

1.50 — Question 15. A teacher plots the scatter diagram below showing the arithmetic and spelling scores from a recent test sat by their students.



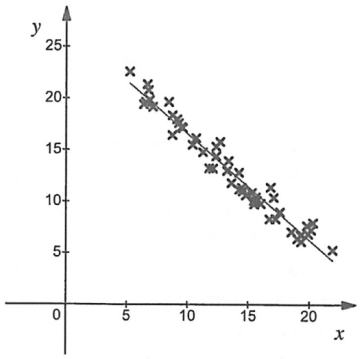
a) Explain why data point A may have been inputted correctly.
b) Explain why data point B may not have been inputted correctly.

1.50 — Question 16. Tristan is doing some research into how sleep cycles affect productivity.
In his trial, he decides to randomly sample twelve adults and when they wake up they must complete a task.
The results of his experiment are shown in the diagram below.



Tristan claims that the scatter diagram proves that increasing the number of hours of sleep reduces the amount of time it takes to complete the task.
Comment on the validity of this claim.

1.50 — Question 17. The diagram below shows bivariate data.



The product moment correlation coefficient for the data is -0.979. Explain why it would be reasonable to model the relationship between x and y as a linear graph.

1.50 — Question 18. Maria wants to compare students' scores from a practice paper, where the maximum score was 100, against their total final exam score, where the maximum score was 300. She takes a random sample of students and the data is shown in the table below.

Score from Practice Paper (/100)	Total Final Exam Score (/300)
72	217
55	168
83	267
46	121
79	202
63	199
88	243
91	265

Use your calculator to calculate the equation of the regression line, where x is the score on the practice paper and y is the total final exam score.

1.50 — Question 19. Chris is investigating how the amount of debt an adult has affects their credit score. He takes a random sample of adults with debts and collects data on the amount of debt they have, in thousands of pounds, and their credit scores. The data is shown in the table below.

Amount of Debt (in thousands of dollars)	Credit Score
12	723
18	716
21	663
27	668
37	672
38	650
41	565
49	575

Use your calculator to calculate the equation of the regression line, where x is the amount of debt and y is the credit score.

1.50 — Question 20. The regression line between a company's monthly advertising expenditure (£a) and its monthly sales revenue (£s) is given by $s = 1.96a + 2920$. Describe in context what the 1.96 represents.

1.50 — Question 21. A company makes cans of soup that are sold at different price points across the country. They want to investigate the relationship between how much the soup costs and how much they sell. The regression line between the price of a can of soup (p pence) and the number of cans sold in a week (N) is given by $N = 502 - 2.22p$. Describe in context what the -2.22 represents.

1.50 — Question 22. Qaisn is investigating how sleep affects sprinters' performance. He takes a random sample of runners and asks them how much sleep they had the night before, in hours, and then records their times running 100 metres. The regression line between the amount of sleep (s hours) and their running times (t seconds) is given by $t = 12.3 - 0.15s$. Describe a limitation with this model.

1.50 — Question 23. Throughout the year, Peter runs a drop-in lunchtime workshop for his students that works on exam questions and exam technique. He maintains a register throughout the year and records the total number of hours each student attended, and their scores in the final exam (which is out of a maximum score of 300). The regression line between the number of hours they attended the workshop (t) and their final exam score (q) is given by $q = 98 + 1.12t$.

a) State the units of the value 1.12 in the regression line equation.

b) With reference to the regression line, describe what effect, on average, attending 2.5 hours of the lunchtime workshop throughout the year would have on a student's final exam score.

The one-hour lunchtime workshop runs Monday to Friday every week throughout the academic year for 38 weeks.

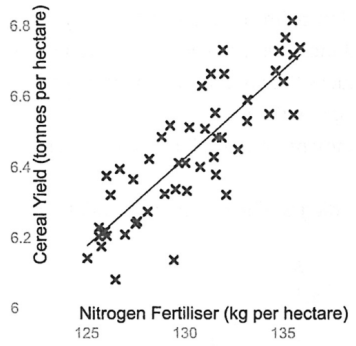
c) Peter wants to use the model to estimate the value of q when $t = 200$. Explain why this would not be appropriate.

1.50 — Question 24. Amelia is investigating the growth of a fungus. She measures the width of the fungus (P mm) every twelve hours (t). She models the growth of the fungus using the regression line given by the equation $P = 3.1 + 0.67t$.

a) With reference to the regression line, describe what effect, on average, three days will have on the width of the fungus.

b) Assuming that the model continues to be accurate, use the regression line to determine an estimate for how many days it would take the fungus to reach a width of 5 cm.

1.50 — Question 25. A random sample of farms report the amount of Nitrogen Fertiliser (kilogram per hectare) they used and their Crop Yield (tonnes per hectare). The data is shown in the diagram below.



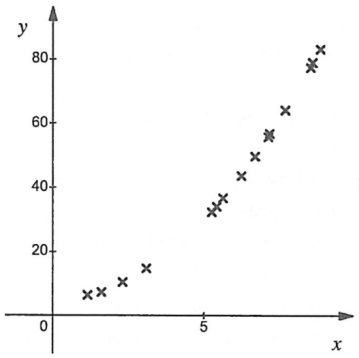
The equation of the regression line is $y = 0.0497x - 0.0277$, where y is the cereal yield and x is the amount of nitrogen fertiliser.

a) Farm A reports that it uses 132 kg per hectare of nitrogen fertiliser. Use the equation of the regression line to estimate Farm A's cereal yield.

b) Farm B reports that it uses 121 kg per hectare of nitrogen fertiliser. Use the equation of the regression line to estimate Farm B's cereal yield.

c) Explain why the estimate for Farm A's cereal yield is more reliable than the estimate for Farm B's cereal yield.

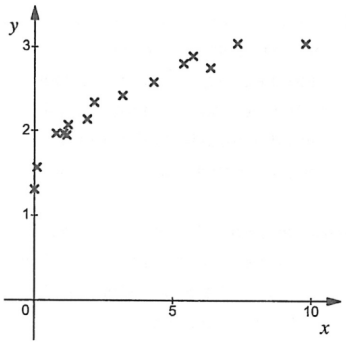
1.51 — Question 1. The diagram below shows bivariate data.



Which of the following best describes the data shown in the diagram?

Positive correlation Positive nonlinear association	Negative correlation Negative nonlinear association

1.51 — Question 2. The diagram below shows bivariate data.



Which of the following best describes the data shown in the diagram?

Positive correlation Positive nonlinear association	Negative correlation Negative nonlinear association

1.51 — Question 3. Barnaby collects the bivariate data shown in the table below.

x	0.8	1.2	2.6	2.9	1.9	1.6	2.5	2.1
y	2.5	1.9	5.8	8.4	4.3	0.9	5.9	4.0

He ranks the data from smallest to largest and correctly finds the differences, d, and the squares of the differences, d^2, as shown in the table below.

Rank x	1	2	7	8	4	3	6	5
Rank y	3	2	6	8	5	1	7	4
d	-2	0	1	0	-1	2	-1	1
d^2	4	0	1	0	1	4	1	1

a) Given that $\sum d_i^2$ is the sum of the squares of the differences, and n is the number of pairs of bivariate data in the original table, calculate Spearman's Rank correlation coefficient using the formula

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

b) Which of the following best describes the association found in a)?

Strong Positive	Moderate Positive
Weak Positive	Weak Negative
Moderate Negative	Strong Negative

1.51 — Question 4. Calculate Spearman's Rank correlation coefficient for the following bivariate data:

x	4.3	3.8	8.6	8.0	1.1	5.5	4.8	6.5
y	7.9	8.1	7.3	7.2	9.3	7.7	7.8	7.6

1.51 — Question 5. Uzair wants to investigate whether there is an association between the number of hours students study, x, and their scores on a maths test, y. The data collected from a random sample of 8 students is shown in the table below.

Student	A	B	C	D	E	F	G	H
x	2	3	4	5	1	6	3	2
y	49	58	62	60	59	71	46	66

Calculate Spearman's Rank correlation coefficient for the data.